The Amdeberhan–Callan Gaussian-product inequality

Principia Math in Progress
MathDB problem 358005

29 September 2026

Abstract

We prove the coefficientwise inequality in Conjecture 15 of Amdeberhan and Callan's final paper *Gaussian inequality*. A Gaussian Vandermonde expansion and the subspace-intersection linearization formula express its difference as a positive combination of two-by-two Gaussian Pascal minors. A subset injection proves the required positivity of these minors.

Write $B(n,r) = \begin{bmatrix} n \\ r \end{bmatrix}_q$, with $B(n,r) = 0$ if $r < 0$ or $r > n$. All upper indices below are nonnegative integers. We use $P \geq_q 0$ to mean $P \in \mathbb{N}[q]$, where zero is included in $\mathbb{N}$.

Theorem 1. *For all integers $0 \leq k \leq a < b$,*

$$D_{a,b,k}(q) := B(a,k)B(a+b,b-k) - B(b,k)B(a+b,a-k) \geq_q 0. \tag{1}$$

Equivalently, the following rational expression is a polynomial with nonnegative integer coefficients:

$$B(a,k)B(b,k)B(a+b,b)\left(\frac{1}{B(a+k,k)} - \frac{1}{B(b+k,k)}\right).$$

The target is Conjecture 4 on page 7 of the arXiv version [2] and Conjecture 15 on page 1533 of the final journal version [1]. We prove the full stated parameter range. The standard identities used below are proved here for completeness.

Lemma 2 (Gaussian Pascal minors)**.** *For $0 \leq a \leq b$ and $0 \leq r \leq s$,*

$$\Delta_{a,b}(r,s) := B(a,r)B(b,s) - B(a,s)B(b,r) \geq_q 0.$$

Proof. The case $r = s$ is zero. If $s > a$, the subtracted product is zero, so the assertion is immediate. Suppose $r < s \leq a$, and put $d = s - r > 0$. Write $[n] = \{1, \ldots, n\}$.

Start with an s-subset I of $[a]$ and an r-subset J of $[b]$. For $0 \leq h \leq a$, set

$$f(h) = |I \cap [h]| - |J \cap [h]|.$$

We have $f(0) = 0$, $f(a) \geq s - r = d$, and each step changes f by at most one. Let h be the first index with $f(h) = d$. Exchange the memberships of I and J on $[h]$:

$$I' = (I \setminus [h]) \cup (J \cap [h]), \qquad J' = (J \setminus [h]) \cup (I \cap [h]).$$

Then $I' \subseteq [a]$, $|I'| = r$, $J' \subseteq [b]$, and $|J'| = s$. In the image, h is the first index for which $|J' \cap [h]| - |I' \cap [h]| = d$. Exchanging the same prefix recovers (I, J), so this is an injection.

It preserves the monomial weight $\prod_{t\in I} x_t \prod_{t\in J} x_t$. Now specialize $x_t = q^{t-1}$. The subset identity

$$\sum_{\substack{S\subseteq[n]\\|S|=\ell}} q^{\sum_{t\in S}(t-1)} = q^{\binom{\ell}{2}} B(n,\ell)$$

shows that the generating polynomial of the unused image-side pairs is $q^{\binom{r}{2}+\binom{s}{2}}\Delta_{a,b}(r,s)$. Every such pair has exponent at least $\binom{r}{2}+\binom{s}{2}$. Removing this common monomial therefore gives a polynomial in $\mathbb{N}[q]$, as required. □

Lemma 3 (Subspace-intersection linearization)**.** *For nonnegative integers n, k, j,*

$$B(n,k)B(n,j) = \sum_{i=0}^{\min(k,j)} q^{(k-i)(j-i)} B(k+j-i,k)B(k,i)B(n,k+j-i). \tag{2}$$

Proof. At a prime power $q = Q$, $B(n,\ell)$ counts the ℓ-dimensional subspaces of an n-dimensional vector space over $\mathbb{F}_Q$. Indeed, counting ordered independent ℓ-tuples and dividing by the number of ordered bases of $\mathbb{F}_Q^\ell$ gives $\prod_{t=0}^{\ell-1}(Q^n - Q^t)/(Q^\ell - Q^t)$, which is $B(n,\ell)$ at $q = Q$.

Count ordered pairs of subspaces U, V with dimensions k, j, according to $i = \dim(U \cap V)$. First choose $W = U + V$, of dimension $k + j - i$; then choose $U \subseteq W$, and then $T = U \cap V \subseteq U$. These choices contribute respectively

$$B(n, k+j-i), \qquad B(k+j-i, k), \qquad B(k,i)$$

evaluated at Q. In W/T, the subspace V/T is a complement to U/T. Fixing one complement, all others are graphs of linear maps from a space of dimension $j-i$ to one of dimension $k-i$. There are $Q^{(k-i)(j-i)}$ such maps. This proves the identity at every prime power. Both sides are polynomials, so agreement at infinitely many values proves (2). If $k > n$ or $j > n$, every term is zero by the convention on B, so these cases are also covered. □

Proof of Theorem 1. The Gaussian Vandermonde identity gives

$$B(a+b, b-k) = \sum_{j\geq 0} q^{j(j+k)} B(a,j)B(b,k+j), \tag{3}$$

$$B(a+b, a-k) = \sum_{j\geq 0} q^{j(j+k)} B(b,j)B(a,k+j). \tag{4}$$

For example, to prove (3) at $q = Q$, count $(b-k)$-subspaces of $X \oplus Y$, where $\dim X = b$ and $\dim Y = a$, by the dimension j of their projection to Y. Choose the intersection with X, of dimension $b-k-j$, and the image in Y. For these choices, the subspaces are graphs of maps from that image to the quotient of X by the intersection, giving $Q^{j(k+j)}$ choices. Polynomial identity and symmetry $B(b, b-k-j) = B(b, k+j)$ prove (3); exchanging a, b gives (4).

Multiply (3) by $B(a,k)$, multiply (4) by $B(b,k)$, and subtract. We obtain

$$D_{a,b,k} = \sum_{j\geq 0} q^{j(j+k)} \big(B(a,k)B(a,j)B(b,k+j) - B(b,k)B(b,j)B(a,k+j)\big).$$

Apply Lemma 3 to the two products having equal upper indices. This yields the explicit expansion

$$\boxed{D_{a,b,k} = \sum_{j=0}^{a} \sum_{i=0}^{\min(k,j)} q^{j(j+k)+(k-i)(j-i)} B(k+j-i,k)B(k,i)\Delta_{a,b}(k+j-i,k+j).} \tag{5}$$

Terms with $j > a$ vanish in the preceding expression: both $B(a, j)$ and $B(a, k+j)$ are zero. Every exponent in (5) is nonnegative, and every factor is in $\mathbb{N}[q]$, using Lemma 2. This proves (1), including $k = 0$, for which all minors in (5) have equal column indices and vanish.

Finally, cancellation of the Gaussian factorials gives

$$\frac{B(a,k)B(b,k)B(a+b,b)}{B(a+k,k)} = B(a,k)B(a+b,b-k).$$

The analogous identity with a, b exchanged gives the second term of (1). This proves the claimed equivalence as an identity of rational functions, whose difference is the polynomial already shown to be nonnegative. □

Scope and verification. No finite cutoff enters the proof. The Pascal-minor injection, subspace-intersection formula, and Vandermonde formula are standard ingredients; the result here is their application through (5) to the full codicil conjecture. The accompanying exact verifier checks the target, both auxiliary identities, the double-sum identity, and the injection itself by exhaustive subset enumeration on small ground sets. The source and bounded prior-claim audit are recorded separately.

Task 2. A direct partition injection. Construct a size-preserving injection directly between the pairs of bounded partitions enumerated by the two products in (1), with an explicit description of the unused pairs.

Task 3. Coefficient shape. Determine whether the nonzero coefficients of $D_{a,b,k}$ are always unimodal. If not, characterize a substantial parameter range where unimodality holds.

Task 4. Support and sharp lower bounds. Determine exactly which coefficients of $D_{a,b,k}$ are positive, and derive uniform lower bounds from selected summands of (5).

Task 5. Further product comparisons. For fixed $a < b$ and k, classify the integers N, u, v for which $B(a,k)B(N,u) - B(b,k)B(N,v)$ is coefficientwise nonnegative. Identify families admitting the same linearization into Pascal minors.

References

[1] T. Amdeberhan and D. Callan, *Gaussian inequality*, Rocky Mountain Journal of Mathematics **54** (2024), no. 6, 1527–1534. doi:10.1216/rmj.2024.54.1527. Conjecture 15, p. 1533.

[2] T. Amdeberhan and D. Callan, *Gaussian inequality*, arXiv:2304.03395v1 (2023). Conjecture 4, p. 7.